

Midterm Report

Start Assignment

- Due Mar 9 by 11:59pm
- Points 100
- Submitting a file upload

For your term project, you need to define a forward problem and devise/implement an inversion strategy to solve the problem. This may mean creating your own inversion code from scratch, or using existing software to run the inversion. If using existing software, since most such software are a "black box", you are expected to demonstrate a clear and thorough understanding of how the software works, and possibly even modify the code to test its behavior (e.g., create and implement a new regularization operator).

For the midterm report, we will focus on the forward problem. Your report should include:

- introduction, motivation and explanation of your term project (~2 pages plus supporting figures)
- physical description of the forward operator, including the mathematical expression itself
- describe the discretization of your model domain
- calculate and plot synthetic forward data for a few simplified models that are representative of your problem
- discuss common sources of noise in real-world data, and mimic such noise by adding it to your synthetic data. Plot a comparison of noise-free and noisy synthetic data
- briefly discuss what kind of inversion strategy is needed to solve your problem (this will be the focus of the final)

You will give a 7-minute presentation on your midterm report detailing each of the bullet points above.